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PROFILE

I am an Assistant Professor at the Department of Computer Science and Engineering, the Hong Kong University of Science and Technology. My main research focuses are large-scale distributed machine learning systems and data management with foundation models.

EXPERIENCE

Assistant Professor, the Hong Kong University of Science and Technology 2023/08 - Present

Department of Computer Science and Engineering

Senior Research Scientist, TogetherAI 2022/10 - 2023/07

Early Stage Technical Member

Postdoc Research Scientist, ETH Zurich 2021/05 - 2023/05

System Group, supervised by Dr. Ce Zhang

EDUCATION

Rice University, Houston, TX, USA 2016/08 - 2020/12

Ph.D., computer science, supervised by Dr. Chris Jermaine

Rice University, Houston, TX, USA 2013/08 - 2016/05

M.S., computer science, supervised by Dr. Ron Goldman

Fudan University, Shanghai, China 2009/09 - 2013/07

B.S., Computer Science

AWARDS

2024 TMLR Outstanding Certification

2020 SIGMOD Research Highlight Award

2019 VLDB Best Paper Honorable Mention Award

2011, 2012 Chinese National Scholarship

SELECTED PUBLICATIONS

- [Haohui Mai](#), Xiaoyan Guo, Xiangyun Ding, [Daifeng Li](#), Qiuchu Yu, Chenzhun Guo, Cong Wang, Jiacheng Zhao, Christos Kozyrakis, and **Binhang Yuan**. "Guiding Agentic GPU Kernel Optimization with Data Flow Invariants." In the 32nd ACM SIGOPS Symposium on Operating Systems Principles. (SOSP 2026)
- [Ding Pan](#), Zhuangzhuang Zhou, Long Qian and **Binhang Yuan**. "Trident: Adaptive Scheduling for Heterogeneous Multimodal Data Pipelines." In Proceedings of the VLDB Endowment, 19(12). (VLDB 2026)
- [Zipeng Qiu](#)*, Wenjie Qu*, Jiaheng Zhang, and **Binhang Yuan**. "V3DB: Audit-on-Demand Zero-Knowledge Proofs for Verifiable Vector Search over Committed Snapshots." In Proceedings of the VLDB Endowment, 19(12). (VLDB 2026)
- [Youhe Jiang](#)*, [Wenshuang Li](#)*, [You Peng](#), Jintao Zhang, [Ran Yan](#), Jianfei Chen, Xu Han, Fangcheng Fu, and **Binhang Yuan**. "HexGen-3: A Fully Disaggregated LLM Serving Framework with Fine-Grained Heterogeneous Resource Autoscaling." In the International Conference on Machine Learning. (ICML 2026)
- Chonghao Zhong, Linfeng Shi, Chen Hua, Tiecheng Sun, Hao Zhao, **Binhang Yuan**, and Chaojian Li. "Scalable Training of 3D Gaussian Splatting via Out-of-Core Optimization." In the International Conference on Machine Learning. (ICML 2026 *Selected as Spotlight*)

- Haobo Lin*, Tianyi Bai*, Chen Chen, Jiajun Zhang, Bohan Zeng, Wentao Zhang, and **Binhang Yuan**. “Synthesizing Multimodal Geometry Datasets from Scratch and Enabling Visual Alignment via Plotting Code” in the International Conference on Machine Learning. (ICML 2026)
- Jiarui Zhang*, Yuchen Yang*, Ran Yan, Zhiyu Mei, Liyuan Zhang, Daifeng Li, Wei Fu, Jiaxuan Gao, Shusheng Xu, Yi Wu, and **Binhang Yuan**. “AREAL-DTA: Dynamic Tree Attention for Efficient Reinforcement Learning of Large Language Models.” In the International Conference on Machine Learning. (ICML 2026)
- Guangxin He*, Yuan Cao*, Yutong He, Tianyi Bai, Kai Chen, Kun Yuan, and **Binhang Yuan**. “TAH-Quant: Effective Activation Quantization in Pipeline Parallelism over Slow Network.” In Transactions on Machine Learning Research, 2026. (TMLR 2026)
- Chuanhao Yan, Fengdi Che, Xuhan Huang, Xu Xu, Xin Li, Yizhi Li, Xingwei Qu, Jingzhe Shi, Chenghua Lin, Yaodong Yang, **Binhang Yuan**, Hang Zhao, Yu Qiao, Bowen Zhou, and Jie Fu. “Re:Form — Reducing Human Priors in Scalable Formal Software Verification with RL in LLMs: A Preliminary Study on Dafny.” In Transactions on Machine Learning Research, 2026. (TMLR 2026)
- Chenyue Li*, Hyeonjae Kim*, Wen Deng, Mengxi Jin, Wen Huang, Mengqian Lu, and **Binhang Yuan**. “ClimateAgent: Multi-Agent Orchestration for Complex Climate Data Science Workflows.” In Transactions on Machine Learning Research, 2026. (TMLR 2026)
- Xiaozhe Yao, Youhe Jiang, Ilia Badanin, Qinghao Hu, **Binhang Yuan**, Imanol Schlag, Eiko Yoneki, and Ana Klimovic. “LLMFabric: Unifying Decentralized HPC Clusters for Heterogeneous LLM Serving.” In 20th USENIX Symposium on Operating Systems Design and Implementation, 2026. (OSDI 2026)
- You Peng*, Youhe Jiang*, Wenqi Jiang, Chen Wang, and **Binhang Yuan**. “Hexgen-Flow: Optimizing LLM Inference Request Scheduling for Agentic Text-to-SQL.” In the 42nd IEEE International Conference on Data Engineering, 2026. (ICDE 2026)
- Ran Yan*, Youhe Jiang*, Xiaonan Nie, Fangcheng Fu, Bin Cui, and **Binhang Yuan**. “HexiScale: Facilitating Large Language Model Training over Heterogeneous Hardware.” To Appear in 9th Annual Conference on Machine Learning and Systems (MLSys 2026)
- Guangxin He, Shen Nie, Fengqi Zhu, Yuankang Zhao, Tianyi Bai, Ran Yan, Jie Fu, Chongxuan Li, and **Binhang Yuan**. “UltraLLaDA: Scaling the Context Length to 128K for Diffusion Large Language Models.” In the 14th International Conference on Learning Representations 2026. (ICLR 2026)
- Di Chai, Pengbo Li, Feiyuan Zhang, Yilun Jin, Han Tian, Kaiqiang Xu, **Binhang Yuan**, Dian Shen, Junxue Zhang, and Kai Chen. “Unlocking Full Efficiency of Token Filtering in Large Language Model Training.” In the 14th International Conference on Learning Representations 2026. (ICLR 2026)
- Ran Yan*, Youhe Jiang*, Zhuoming Chen, Haohui Mai, Beidi Chen, and **Binhang Yuan**. “FSA: An Alternative Efficient Implementation of Native Sparse Attention Kernel.” In the 14th International Conference on Learning Representations 2026. (ICLR 2026)
- Youhe Jiang*, Fangcheng Fu*, Wanru Zhao*, Stephan Rabanser, Jintao Zhang, Nicholas D. Lane, and **Binhang Yuan**. “Cascadia: An Efficient Cascade Serving System for Large Language Models.” In the 14th International Conference on Learning Representations 2026. (ICLR 2026)
- Lingfei Zeng, Fengdi Che, Xuhan Huang, Fei Ye, Xu Xu, **Binhang Yuan**, and Jie Fu. “VeriEquivBench: An Equivalence Score for Ground-Truth-Free Evaluation of Formally.” In the 14th International Conference on Learning Representations 2026. (ICLR 2026)
- Fangyu Ding, Ding Ding, Sijin Chen, Kaibo Wang, Peng Xu, Zijin Feng, Haoli Bai, Kai Han, Youliang Yan, **Binhang Yuan**, and Jiacheng Sun. “Beyond Masks: Efficient, Flexible Diffusion Language Models via Deletion-Insertion Processes.” In the 14th International Conference on Learning Representations 2026. (ICLR 2026)
- Xu Xu*, Xin Li*, Xingwei Qu, Jie Fu, and **Binhang Yuan**. “Local Success Does Not Compose: Benchmarking Large Language Models for Compositional Formal Verification.” In the 14th International Conference on Learning Representations 2026. (ICLR 2026)
- Wei Fu, Jiaxuan Gao, Xujie Shen, Chen Zhu, Zhiyu Mei, Chuyi He, Shusheng Xu, Guo Wei, Jun Mei, Jiashu Wang, Tongkai Yang, **Binhang Yuan**, and Yi Wu. “AREAL: Asynchronous Reinforcement Learning for Efficient and Scalable Language Reasoning.” In Advances in Neural Information Processing Systems 39 (2025). (NeurIPS 2025)
- Chenyue Li, Wen Deng, Mengqian Lu, and **Binhang Yuan**. “AtmosSci-Bench: Evaluating the Recent Advance of Large Language Model for Atmospheric Science” In Advances in Neural Information Processing Systems 39 (2025). (NeurIPS 2025)
- Tianyi Bai, Zengjie Hu, Fupeng Sun, Qiu Jiantao, Yizhen Jiang, Guangxin He, Bohan Zeng, Conghui He, **Binhang Yuan**, and Wentao Zhang. “Multi-step Visual Reasoning with Visual Tokens Scaling and Verification.” In Advances in Neural Information Processing Systems 39 (2025). (NeurIPS 2025)

- Tianyi Bai, Yuxuan Fan, Qiu Jiantao, Fupeng Sun, Jiayi Song, Junlin Han, Zichen Liu, Conghui He, Wentao Zhang, and **Binhang Yuan**. "Hallucination at a Glance: Controlled Visual Edits and Fine-Grained Multimodal Learning." In Advances in Neural Information Processing Systems 39 (2025). (NeurIPS 2025)
- Guoliang He, Youhe Jiang, Wencong Xiao, Jiang Kaihua, Shuguang Wang, Jun Wang, Du Zixian, Zhuo Jiang, Xinlei Zhang, **Binhang Yuan**, and Eiko Yoneki. "Efficient Pre-Training of LLMs via Topology-Aware Communication Alignment on More Than 9600 GPUs." In Advances in Neural Information Processing Systems 39 (2025). (NeurIPS 2025)
- Rizhen Hu, Yutong He, Ran Yan, Mou Sun, **Binhang Yuan**, and Kun Yuan. "MeCeFO: Enhancing LLM Training Robustness via Fault-Tolerant Optimization." In Advances in Neural Information Processing Systems 39 (2025). (NeurIPS 2025)
- Tianyi Bai, Ling Yang, Zhenhao Wong, Fupeng Sun, Xinlin Zhuang, Jiahui Peng, Chi Zhang, Lijun Wu, Jiantao Qiu, Wentao Zhang, **Binhang Yuan**, and Conghui He. "Efficient Pretraining Data Selection for Language Models via Multi-Actor Collaboration." In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, pp. 9465-9491. (ACL 2025).
- Youhe Jiang, Fangcheng Fu, Xiaozhe Yao, Guoliang HE, Xupeng Miao, Ana Klimovic, Bin Cui, **Binhang Yuan**, and Eiko Yoneki. "Demystifying Cost-Efficiency in LLM Serving over Heterogeneous GPUs." In the International Conference on Machine Learning. (ICML 2025)
- Suyi Li, Hanfeng Lu, Tianyuan Wu, Minchen Yu, Qizhen Weng, Xusheng Chen, Yizhou Shan, **Binhang Yuan**, and Wei Wang. "Toppings: CPU-Assisted, Rank-Aware Adapter Serving for LLM Inference." To Appear in the 2025 USENIX Annual Technical Conference (ATC 2025).
- Wenjie Qu*, Yuguang Zhou*, Yongji Wu, Tingsong Xiao, **Binhang Yuan**, Yiming Li, and Jiaheng Zhang. "Prompt Inversion Attack against Collaborative Inference of Large Language Models." In IEEE Symposium on Security and Privacy, pp. 1695-1712, 2025 (S&P 2025).
- Jinwei Yao, Kaiqi Chen, Kexun Zhang, Jiaxuan You, **Binhang Yuan**, Zeke Wang, Tao Lin. "DeFT: Decoding with Flash Tree-attention for Efficient Tree-structured LLM Inference" In the 13th International Conference on Learning Representations 2025. (ICLR 2025 *Selected as Spotlight*)
- Youhe Jiang*, Ran Yan*, and **Binhang Yuan**. "HexGen-2: Disaggregated Generative Inference of LLMs in Heterogeneous Environment" In the 13th International Conference on Learning Representations 2025. (ICLR 2025)
- Changyue Liao, Mo Sun, Zihan Yang, Jun Xie, Kaiqi Chen, **Binhang Yuan**, Fei Wu, and Zeke Wang. "Ratel: Optimizing Holistic Data Movement to Fine-tune 100B Model on a Consumer GPU." In the 41st IEEE International Conference on Data Engineering, pp. 292-306, 2025. (ICDE 2025)
- Yongjun He, Roger Waleffe, Zhichao Han, Johnu George, **Binhang Yuan**, Zitao Zhang, Yinan Shan, Yang Zhao, Debojyoti Dutta, Theodoros Rekatsinas, and Ce Zhang. "MLKV: Efficiently Scaling up Large Embedding Model Training with Disk-based Key-Value Storage." In the 41st IEEE International Conference on Data Engineering, pp. 4134-4141. 2025. (ICDE 2025)
- Xinyu Zhao*, Guoheng Sun*, Ruisi Cai*, Yukun Zhou*, Pingzhi Li*, Peihao Wang, Bowen Tan, Yexiao He, Li Chen, Yi Liang, Beidi Chen, **Binhang Yuan**, Hongyi Wang, Ang Li, Zhangyang Wang, Tianlong Chen. "Model-Glue: Democratized LLM Scaling for A Large Model Zoo in the Wild." In Advances in Neural Information Processing Systems 38 (2024). (NeurIPS 2024)
- Youhe Jiang*, Ran Yan*, Xiaozhe Yao*, Yang Zhou, Beidi Chen, and **Binhang Yuan**. "HexGen: Generative Inference of Large-Scale Foundation Model over Heterogeneous Decentralized Environment." In International Conference on Machine Learning (pp. 21946-21961). PMLR. (ICML 2024)
- Lin Lu*, Chenxi Dai*, Wangcheng Tao, **Binhang Yuan**, Yanan Sun, and Pan Zhou "Exploring the Robustness of Pipeline-Parallelism-Based Decentralized Training." In International Conference on Machine Learning (pp. 32978-32989). PMLR. (ICML 2024)
- Alexandre E Eichenberger, Qi Lin, Saif Masood; Hong Min, Alex Sim, Yida Wang, Kesheng Wu, **Binhang Yuan**, Lixi Zhou, and Jia Zou. "Serving Deep Learning Model in Relational Databases." In 27th International Conference on Extending Database Technology 2024. EDBT, pp. 717-724. (EDBT 2024)
- Ying Sheng, Lianmin Zheng, **Binhang Yuan**, Zhuohan Li, Max Ryabinin, Daniel Y. Fu, Zhiqiang Xie, Beidi Chen, Clark Barrett, Joseph E. Gonzalez, Percy Liang, Christopher Ré, Ion Stoica, and Ce Zhang. "High-throughput Generative Inference of Large Language Models with a Single GPU." In International Conference on Machine Learning (pp. 31094-31116). PMLR. (ICML 2023 *Selected as Oral*).
- Zichang Liu, Jue Wang, Tri Dao, Tianyi Zhou, **Binhang Yuan**, Zhao Song, Anshumali Shrivastava, Ce Zhang, Yuandong Tian, Christopher Ré, and Beidi Chen. "Deja Vu: Contextual Sparsity for Efficient LLMs at Inference Time." In International Conference on Machine Learning (pp. 22137-22176). PMLR. (ICML 2023 *Selected as Oral*).

- Jue Wang, Yucheng Lu, **Binhang Yuan**, Beidi Chen, Percy Liang, Christopher De Sa, Christopher Ré, and Ce Zhang. "CocktailSGD: Fine-tuning Foundation Models over 500Mbps Networks." In International Conference on Machine Learning (pp. 36058-36076). PMLR. (ICML 2023)
- Yuxin Tang, Zhimin Ding, Dimitrije Jankov, **Binhang Yuan**, Daniel Bourgeois, and Chris Jermaine. "Auto-Differentiation of Relational Computations for Very Large Scale Machine Learning." In International Conference on Machine Learning (pp. 33581-33598). PMLR. (ICML 2023)
- **Binhang Yuan***, Yongjun He*, Jared Quincy Davis, Tianyi Zhang, Tri Dao, Beidi Chen, Percy Liang, Christopher Re, and Ce Zhang. "Decentralized Training of Foundation Models in Heterogeneous Environments." In Advances in Neural Information Processing Systems 36 (2022), 25464-25477. (NeurIPS 2022 *Selected as Oral*)
- Jue Wang*, **Binhang Yuan***, Luka Rimanic*, Yongjun He, Tri Dao, Beidi Chen, Christopher Re, and Ce Zhang. "Fine-tuning Language Models over Slow Networks using Activation Compression with Guarantees." In Advances in Neural Information Processing Systems 36 (2022), 19215-19230. (NeurIPS 2022)
- Rui Pan, Yiming Lei, Jialong Li, Zhiqiang Xie, **Binhang Yuan**, and Yiting Xia. "Efficient Flow Scheduling in Distributed Deep Learning Training with Echelon Formation." In Proceedings of the twenty first ACM Workshop on Hot Topics in Networks (2022). (HotNets 2022)
- Xiangru Lian, **Binhang Yuan**, Xuefeng Zhu, Yulong Wang, Yongjun He, Honghuan Wu, Lei Sun, Haodong Lyu, Chengjun Liu, Xing Dong, Yiqiao Liao, Mingnan Luo, Congfei Zhang, Jingru Xie, Haonan Li, Lei Chen, Renjie Huang, Jianying Lin, Chengchun Shu, Xuezhong Qiu, Zhishan Liu, Dongying Kong, Lei Yuan, Hai Yu, Sen Yang, Ce Zhang, and Ji Liu. "Persia: An Open, Hybrid System Scaling Deep Learning-based Recommenders up to 100 Trillion Parameters." In Proceedings of the 28th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 3288-3298. 2022 (SIGKDD 2022)
- Lijie Xu, Shuang Qiu, **Binhang Yuan**, Jiawei Jiang, Cedric Renggli, Shaoduo Gan, Kaan Kara, Guoliang Li, Ji Liu, Wentao Wu, Jieping Ye, and Ce Zhang. "In-Database Machine Learning with CorgiPile: Stochastic Gradient Descent without Full Data Shuffle." In Proceedings of the 2022 International Conference on Management of Data, pp. 1286-1300. 2022. (SIGMOD 2022)
- **Binhang Yuan**, Cameron R. Wolfe, Chen Dun, Yuxin Tang, Anastasios Kyrillidis, and Chris Jermaine. "Distributed Learning of Deep Neural Networks using Independent Subnet Training." In Proceedings of the VLDB Endowment, 15(8). (VLDB 2022)
- Shaoduo Gan, Xiangru Lian, Rui Wang, Jianbin Chang, Chengjun Liu, Hongmei Shi, Shengzhuo Zhang, Xianghong Li, Tengxu Sun, Jiawei Jiang, **Binhang Yuan**, Sen Yang, Ji Liu, and Ce Zhang. "BAGUA: Scaling up Distributed Learning with System Relaxations." In Proceedings of the VLDB Endowment, 15(4). (VLDB 2022)
- Shangyu Luo, Dimitrije Jankov, **Binhang Yuan**, and Chris Jermaine. "Automatic Optimization of Matrix Implementations for Distributed Machine Learning and Linear Algebra." In Proceedings of the 2021 International Conference on Management of Data(pp. 1222-1234). ACM. (SIGMOD 2021)
- **Binhang Yuan**, Dimitrije Jankov, Jia Zou, Yuxin Tang, Daniel Bourgeois, and Chris Jermaine. "Tensor Relational Algebra for Machine Learning System Design." In Proceedings of the VLDB Endowment, 14(8), 1338-1350 (VLDB 2021)
- Jia Zou, Pratik Barhate, Amitabh Das, Arun Iyengar, **Binhang Yuan**, Dimitrije Jankov, and Chris Jermaine. "Lachesis: Automatic Partitioning for UDF-Centric Analytics." In Proceedings of the VLDB Endowment, 14(8), 1262-1275 (VLDB 2021)
- Dimitrije Jankov, **Binhang Yuan**, Shangyu Luo, and Chris Jermaine. "Distributed Numerical and Machine Learning Computations via Two-Phase Execution of Aggregated Join Trees." In Proceeding of VLDB Endowment, 14(7), 1228-1240. (VLDB 2021)
- Dimitrije Jankov, Shangyu Luo, **Binhang Yuan**, Zhuhua Cai, Jia Zou, Chris Jermaine, and Zekai J Gao. "Declarative recursive computation on an RDBMS: or, why you should use a database for distributed machine learning." In Proceedings of the VLDB Endowment, 12(7), 822-835. (VLDB 2019, *VLDB Best Paper Honorable Mention Award*)
- Jia Zou, R Matthew Barnett, Tania Lorigo-Botran, Shangyu Luo, Carlos Monroy, Sourav Sikdar, Kia Teymourian, **Binhang Yuan**, and Chris Jermaine. "PlinyCompute: A platform for high-performance, distributed, data-intensive tool development." In Proceedings of the 2018 International Conference on Management of Data(pp. 1189-1204). ACM. (SIGMOD 2018)
- **Binhang Yuan**, Vijayaraghavan Murali, and Chris Jermaine. "Abridging source code." In Proceedings of the ACM on Programming Languages 1.OOPSLA (2017): 58. (OOPSLA 2017)
- Bo Yan, **Binhang Yuan**, and Bo Yang. "Effective video retargeting with jittery assessment." In IEEE Transactions on Multimedia, Vol. 16, Issue 1, pp. 272-277, Jan. 2014. (TMM 2014)

ACADEMIC SERVICE

- Conference reviewer:

- AAAI: 2020, 2021;
- ICLR: 2022, 2023, 2024, 2025;
- ICML: 2021, 2022, 2023, 2024, 2025, 2026;
- NeurIPS: 2020, 2021, 2022, 2023;
- MLSys: 2024, 2025, 2026.
- Conference area chair:
 - NeurIPS: 2024, 2025, 2026;
 - ICLR: 2026, 2027.
- Journal reviewer:
 - IEEE Access: 2020;
 - IEEE TKDE: 2022, 2023;
 - IEEE BigData: 2023;
 - JMLR: 2023;
 - PVLDB: 2023.

OTHER EXPERIENCE

Academic Consulting, Ant Group, Hangzhou, China

2026/01 - 2026/12

Research Intern, Microsoft Research Asia, Beijing, China

2017/07 - 2017/12

SDE Intern, Tableau Software, Seattle, WA, USA

2016/05 - 2016/08

SDE Intern, Isilon EMC2, Santa Clara, CA, USA

2015/05 - 2015/08